

BST Neutrino Sector — Falsifier Data-Sheet (v1, 2026-08-14)

Empirical companion to the BST neutrino theory (Lyra T2563 / F985 spine)

Grace (data lane). Every measurement pulled fresh 2026-08-14 with source + date; live numbers go stale in weeks, so this sheet is dated and versioned. BST predictions are zero-parameter, from D_{IV}^5 (five integers rank=2, $N_c=3$, $n_C=5$, $C_2=6$, $g=7$; $N_{max}=137$). Tier per prediction; distinctive-vs-generic separated explicitly.

The theory in one line (Lyra’s spine)

No right-handed neutrino (forbidden — non-unitary shadow; “no steriles” is a Five-Absence) $\rightarrow$ Majorana $\rightarrow$ effective mass matrix has rank 2 = rank(D_{IV}^5) $\rightarrow$ one-dimensional kernel $\rightarrow m_1 = 0$ exactly, normal ordering. Everything below follows from that rank-2 kernel, coupled to the no-crossing dark-energy result.

1. Neutrino mass sum Σm_ν — the sharpest live kill

	value	source (fresh 2026-08-14)
BST prediction	$\Sigma m_\nu \approx 58 \text{ meV}$ ($m_1=0$; $m_2=\sqrt{\Delta m_{21}^2} \approx 8.6$, $m_3=\sqrt{\Delta m_{31}^2} \approx 50 \text{ meV}$)	F985 / T2197, zero-param
DESI DR2 BAO + Planck + ACT, Λ CDM (95%)	$< 64 \text{ meV}$	arXiv:2503.14744
DESI + CMB, Feldman–Cousins boundary-corrected (95%)	$< 53 \text{ meV}$	same
Oscillation floor (normal ordering)	$> 59 \text{ meV}$	global fits
Ordering evidence (2026)	robust preference for NORMAL	arXiv:2601.16165

- Status: BST’s 58 meV sits $\sim 6 \text{ meV}$ under the Λ CDM bound (SAFE by a hair) and slightly under the aggressive FC bound (53 meV) — but the FC bound breaches the NO oscillation floor (59 meV) too, so that tension is with the WHOLE minimal-NO scenario, not BST-specific.
- Kill condition: a robust Σm_ν measurement $< \sim 58 \text{ meV}$ under no-crossing/ Λ CDM refutes BST (see coupling, Sec 5).
- Favorable: the 2026 normal-ordering preference matches BST’s forced NO.

- Distinctive? NO — the sum ≈ 58 meV is the generic minimal-NO floor (any $m_1=0$ NO theory gives it). Tier I (the $m_1=0$ floor). The BST-specific content is $m_1=0$ *exactly* + the coupling (Sec 5) + the $0\nu\beta\beta$ window (Sec 2).

2. Neutrinoless double-beta decay $m_{\beta\beta}$ — the distinctive kill-test we own

	value	source (fresh 2026-08-14)
BST prediction	$m_{\beta\beta} \in [1.5, 3.7]$ meV	F413/K673, $m_1=0$ -exact (canonical PMNS)
KamLAND-Zen complete dataset (90% CL)	< 28 – 122 meV (NME-dependent); $T_{1/2} > 3.8 \times 10^{26}$ yr	arXiv:2406.11438, PRL 135 (Dec 2025)
Next-gen reach (LEGEND-1000 / nEXO)	~ 10 – 20 meV	design

- Status: BST's 1.5 – 3.7 meV is ~ 10 – $80\times$ BELOW current sensitivity $\rightarrow$ no tension; a null result does NOT refute BST. [Lower edge = $|s_{12}^2 c_{13}^2 \cdot m_2 - s_{13}^2 \cdot m_3| = 1.49$ – 1.51 meV with canonical $\sin^2\theta_{12}=42/137$; corroborated by NuFIT-6.0 central values (1.499). The old 1.44 used the superseded $\sin^2\theta_{12}=3/10$.]
- ★ FORWARD KILL-TEST (two-tier — the ≥ 15 meV IO floor is the excursion-proof kill): BST predicts LEGEND-1000/nEXO will NOT detect $0\nu\beta\beta$.
 - Prediction: $m_{\beta\beta} \in [1.5, 3.7]$ meV (canonical central).
 - Tension (4 – 15 meV): a detection here is in tension but NOT a clean kill — BST's own band reaches ~ 4.2 meV at joint- 3σ (oscillation-parameter excursion) and NME uncertainty spans this range. Nitpickable, not decisive.
 - ★ Clean kill ($\geq \sim 15$ meV, conservative): the inverted-ordering band is ~ 18 – 48 meV (canonical; softening to ~ 15 meV only at the 3σ -soft edge, $s_{12}^2=0.34$). A detection at/above ~ 15 meV falsifies BST — BST forces normal ordering, so an IO-floor signal is excursion-proof. $\geq \sim 15$ meV is the conservative threshold (safely below the ~ 18 meV central IO min and well above BST's NO band, which tops out ~ 4.2 meV); a detection in the ~ 18 – 48 meV band is unambiguously inverted ordering.
- Distinctive? YES — the $[1.5, 3.7]$ meV window + $m_1=0$ -exact is BST-specific (not the generic floor). Also the kill-test for the color-duality theorem T2535 (K1021): if $0\nu\beta\beta$ is NEVER seen AND neutrinos turn out Dirac, the large-PMNS/small-CKM explanation is refuted.

3. Atmospheric mixing θ_{23} — the distinctive octant kill-test

	value	source (fresh 2026-08-14)
BST prediction	$\sin^2\theta_{23} = 4/7 = 0.571$ (UPPER octant)	canonical (Shilov- $\mathbb{Z}_2$ μ - τ , T2534); tier I
NuFIT-6.0 upper octant	$0.561 \pm 0.013 \rightarrow 0.80\sigma$	arXiv:2410.05380
NuFIT-6.0 lower octant	$0.470 \pm 0.015 \rightarrow 6.8\sigma$	same; octant UNRESOLVED

- ★ FORWARD KILL-TEST (distinctive): BST predicts the upper octant; the data is genuinely octant-ambiguous now. A decisively-settled LOWER octant KILLS BST ($\sim 6.8\sigma$). Form-independent — every BST upper-lean form dies. JUNO / DUNE / T2HK resolve the octant (~ 2027 – 2035).
- Distinctive? YES — the octant is a specific, falsifiable, currently-unresolved prediction (the rational $4/7$ itself is consistent-not-sold, tier I).

4. Other PMNS angles + CP phase (consistency, not sold as teeth)

observable	BST (canonical)	value	NuFIT-6.0	σ	tier
$\sin^2\theta_{12}$	C_2·g/N_max = 42/137	0.3066	0.307 ± 0.012	0.04σ	I (most target-innocent — rides forced cap 137)
$\sin^2\theta_{13}$	tie: N_c ² ·n_C=1/45 & N_c/N_max=3/137	0.0222/0.0219	0.0222 ± 0.0006	$0.04\sigma/0.50\sigma$	I (K1260 tie, mechanism-unbroken)
δ_{CP}	$\delta_{\text{PMNS}} \approx 197^\circ$	—	NO: CP-conserving within 1σ	consistent	existence-only (magnitude reverse-fit K1304 — NOT dressed)

- Honest: the individual rational forms are consistent but not sold (any one is a low-denom fit; two forms per angle = IDENTIFIED not forced). The sold teeth are the θ_{23} octant (Sec 3) and the $0\nu\beta\beta$ window (Sec 2). δ_{CP} is existence-only.

5. The coupling — $\text{DE} \leftrightarrow \Sigma m_\nu$ (BST-specific, removes the escape hatch)

BST’s dark-energy prediction is no phantom crossing ($w > -1$, complete-monotonicity F799; pred_125, survives DESI DR2 radial BAO + DES-SN5YR). The Σm_ν cosmology bound is DE-model-dependent: phantom ($w < -1$) LOOSENS it to ~ 0.16 eV; Λ CDM/quintessence keeps it tight. BST’s no-crossing is on the TIGHT side $\rightarrow$ BST cannot retreat to the dynamical-DE relaxation to save Σm_ν . Most $m_1=0$ NO theories can hide in dynamical DE; BST cannot. This welds the neutrino and dark-energy sectors into one coupled exposure — a genuinely BST-specific test.

6. Summary — what falsifies BST’s neutrino sector

test	kill condition	when	distinctive
Σm_ν	robust $< \sim 58$ meV under no-crossing/ Λ CDM	now–2027	coupling, not sum
$0\nu\beta\beta$	clean kill = detection $\geq \sim 15$ meV (conservative; IO band ~ 18 – 48 meV, NO forced). 4–15 meV = tension, not decisive	LEGEND-1000/nEXO, ~ 2030 s	YES
θ_{23} octant	decisively-settled LOWER octant	JUNO/DUNE/T2HK YES ~ 2027 – 2035	
ordering	inverted ordering confirmed	JUNO ~ 2030	YES (NO forced)

The through-line (with Lyra’s theory): the operator has one ground state; the neutrino has one massless species — the same rank-2 asymmetry, “the breaking is the content,” in two sectors.

Sources: NuFit-6.0 arXiv:2410.05380; DESI DR2 neutrino arXiv:2503.14744; DESI DR1 NO reanalysis arXiv:2601.16165; KamLAND-Zen arXiv:2406.11438, PRL 135 (2025). All pulled fresh 2026-08-14. Δm^2 ratio stays OPEN (undressed).